GERARD

Hardware Bible

Hardware revision A: parts, circuits, wiring and enclosure

Release	Version 1 (tag `v.1`), declared 2026-11-09
Hardware revision	A
Firmware version	`v.1.0.20261109T161718Z`
This PDF exported	2026-11-09
Author	Zeroshin

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Colophon

Item	Value
Release	Version 1, declared 2026-11-09
Exported	2026-11-09
Hardware revision	A
Firmware version	`v.1.0.20261109T161718Z`
Author	Zeroshin
Licences	This document: `CC-BY-SA-4.0`. Hardware design (KiCad and STL files): `CERN-OHL-S-2.0`. Firmware: `GPL-3.0-or-later`.
Root repository commit	`83c3228`

Callouts in this document:

> NOTICE. Risk of damage to GERARD or to what it is plugged into, and how to avoid it.

> Note. A fact not to miss. It is not a hazard.

The other book, the Markdown editions of both books and every source file are in the GERARD repository: start with its `README.md`.

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1 GERARD

Figure 1. GERARD, front. The display shows the menu.



GERARD is a desk focus timer. It counts focus time, counts down a set time, and reminds the person at the desk to take breaks, drink and stretch (Figure 1). It shows its screens on a small white OLED display. One rotary knob, which is also a push button, works every screen. A passive magnetic buzzer plays its tunes. It runs from 5 V DC through a USB-C socket on its 3D-printed enclosure.

Inside, an RP2040-Zero board (U1) drives the display (U2), reads the rotary encoder (U3) and switches the buzzer (BZ1) through a transistor (Q1). U1 and the buzzer's driver sit on one perfboard. Two cables join it to the display and the encoder. All parts are through-hole.

Table 1. Where each source file is.

Source file	Where in the repository
Schematic, KiCad 9	`hardware/kicad/gerard.kicad_sch`
Schematic, PDF and SVG	`hardware/kicad/gerard - Schematic.pdf`, `docs/figures/gerard-schematic.svg`
Cable drawing	`docs/figures/gerard-cables.svg`, from `gerard-cables.yml`
Enclosure, STL	`hardware/3d/GERARD top.stl`, `hardware/3d/GERARD bottom.stl`
Knob, STL	`hardware/3d/Knob FINAL.stl`
Firmware	`firmware/`, described in the Firmware Gospel

2 Safety

GERARD runs only from 5 V DC, through the USB-C socket on its enclosure, J9. J9 passes the supply straight through to U1′s own USB-C socket by the plug P1.

> NOTICE. R1, the buzzer's series resistor, must be at least 100 Ω and rated at least 0.25 W. A weaker R1 may not survive the BUZZER line stuck high with Q1 on.

3 Parts

Table 2. Parts.

Ref.	Function	Part and rating	Value	Unit	Polarity
U1	Microcontroller board	RP2040-Zero: RP2040 chip, revision B2, 2048 KB flash, USB-C socket			By the printed pad names
U2	Display	OLED module, all white, I2C, SSD1306 driver; 0.96 inch, 128 x 64 on the schematic			Pins GND, VDD, SCK, SDA
U3	Knob and push button	KY-040 rotary encoder module with push switch			By the printed pin names
BZ1	Buzzer	Passive magnetic buzzer; coil resistance	15.6	Ω	+ and − leads
Q1	Buzzer switch	BC337 NPN transistor: 45 V collector-emitter, 800 mA continuous collector current			C, B, E on the parts fitted
D1	Flyback diode	1N5819 Schottky rectifier: 40 V peak repetitive reverse voltage, 1.0 A average forward current			Band (cathode) on BZ1 +
R1	Buzzer series resistor	Through-hole resistor, 0.25 W or more	100	Ω	None
R2	Base resistor of Q1	Through-hole resistor	1	kΩ	None
R3	Pull-down on Q1′s base	Through-hole resistor	10	kΩ	None
J1	Display cable header	JST XH, 4 ways, 2.5 mm pitch, on the perfboard			By signal
J2	Display cable housing	JST XH, 4 ways, U1′s end of the display cable			By signal
J5	Encoder cable header	JST XH, 5 ways, 2.5 mm pitch, on the perfboard			By signal
J6	Encoder cable housing	JST XH, 5 ways, U1′s end of the encoder cable			By signal
J9	USB-C socket on the enclosure	USB-C female breakout board, 4 holes printed VCC, D-, D+, GND			Holes by name
P1	USB-C plug in U1′s socket	USB-C male plug board, "3P 2P Charging", solder type			By signal (Table 3)
—	Board	One perfboard for U1, BZ1, Q1, D1 and R1 to R3			—
—	Enclosure and knob	Printed in PLA or any material that suits			—

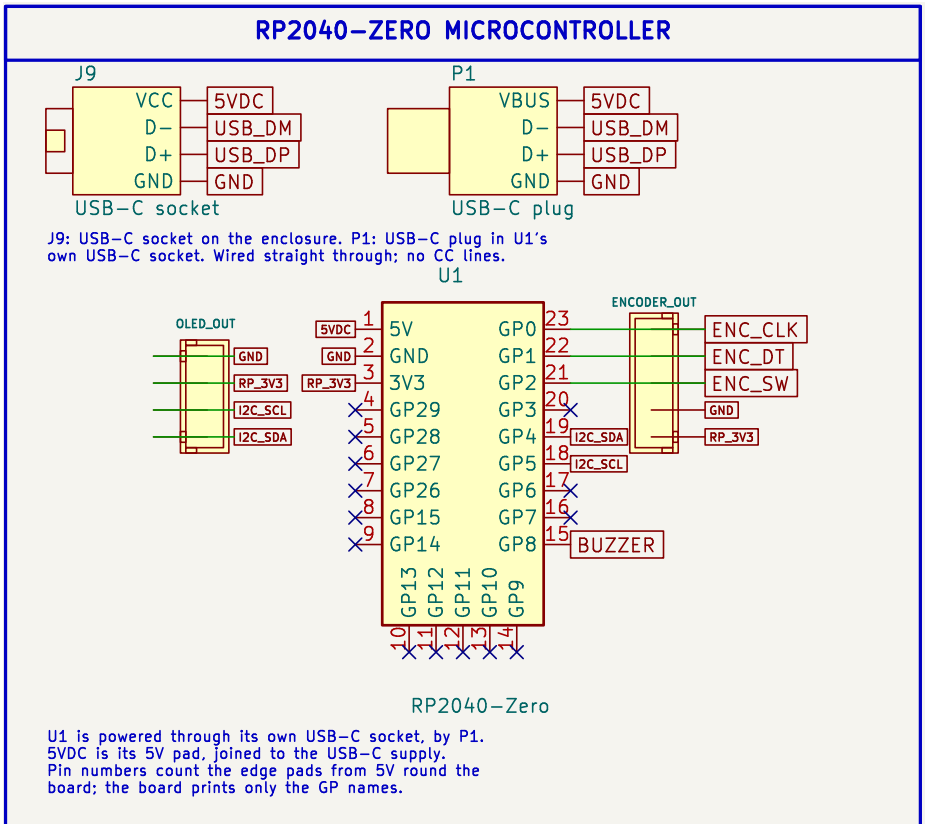
> Note. A BC337′s lead order differs between makers. Find Q1′s collector, base and emitter with a tester before you solder it.

4 Circuits

The schematic, `hardware/kicad/gerard.kicad_sch`, title GERARD, rev A, has four boxes. Each section below shows one box. The whole sheet is in Section 8, page 8.

4.1 Microcontroller and USB-C

Figure 2. Schematic box RP2040-ZERO MICROCONTROLLER.



GERARD's USB-C cable plugs into J9, on the enclosure. J9 is wired straight through to P1, a plug that sits in U1's own USB-C socket (Figure 2). This is easier to manage than a cable plugged into U1 itself. There is no CC line: none is needed, and J9's board brings none out.

CC is the configuration channel of a USB-C connector.

Table 3. USB-C extension, J9 to P1.

Signal	J9 hole	P1 pad
5VDC	VCC	VBUS
USB_DM	D-	D-
USB_DP	D+	D+
GND	GND	GND

U1's 5V pad carries `5VDC`, joined to its USB-C socket's supply. Its GND pad carries `GND`. Its 3V3 pad carries `RP_3V3`, which supplies the display and the encoder.

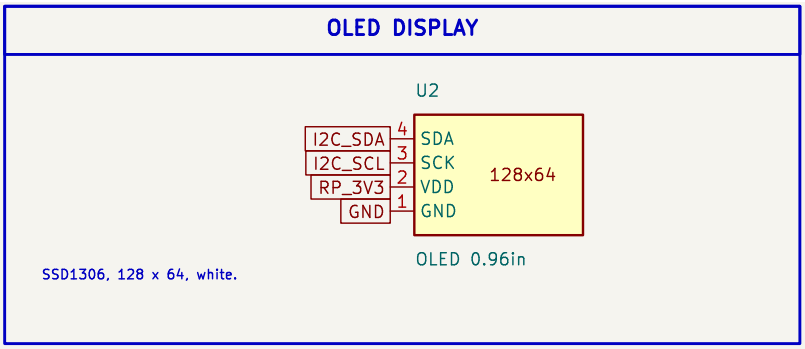
Table 4. U1′s pads that carry signals. The other edge pads are unused.

Signal	U1 pad	RP2040 function	Goes to
ENC_CLK	GP0	GPIO0	U3, pin CLK
ENC_DT	GP1	GPIO1	U3, pin DT
ENC_SW	GP2	GPIO2	U3, pin SW (push button)
I2C_SDA	GP4	I2C0 SDA	U2, pin SDA
I2C_SCL	GP5	I2C0 SCL	U2, pin SCK
BUZZER	GP8	PWM4 A	Through R2 to Q1′s base

The schematic numbers U1′s edge pads from the 5V pad round the board: 5V 1, GND 2, 3V3 3, GP8 15, GP5 18, GP4 19, GP2 21, GP1 22, GP0 23. The board prints only the names.

4.2 Display

Figure 3. Schematic box OLED DISPLAY.

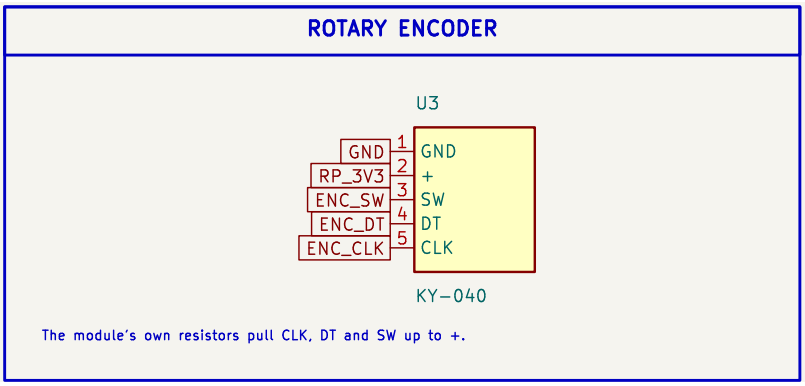


U2′s pins, as printed, run GND, VDD, SCK, SDA (Figure 3). VDD is on `RP_3V3`, SCK on `I2C_SCL`, SDA on `I2C_SDA`. U2 answers on I2C at address 0x3C. It is fitted as it comes, with no pull-up resistor added on its bus.

> Note. An earlier yellow-and-blue module broke and was replaced by the all-white one. Its pins ran VCC, GND, SCL, SDA. Wire any module by the names printed on it.

4.3 Rotary encoder

Figure 4. Schematic box ROTARY ENCODER.

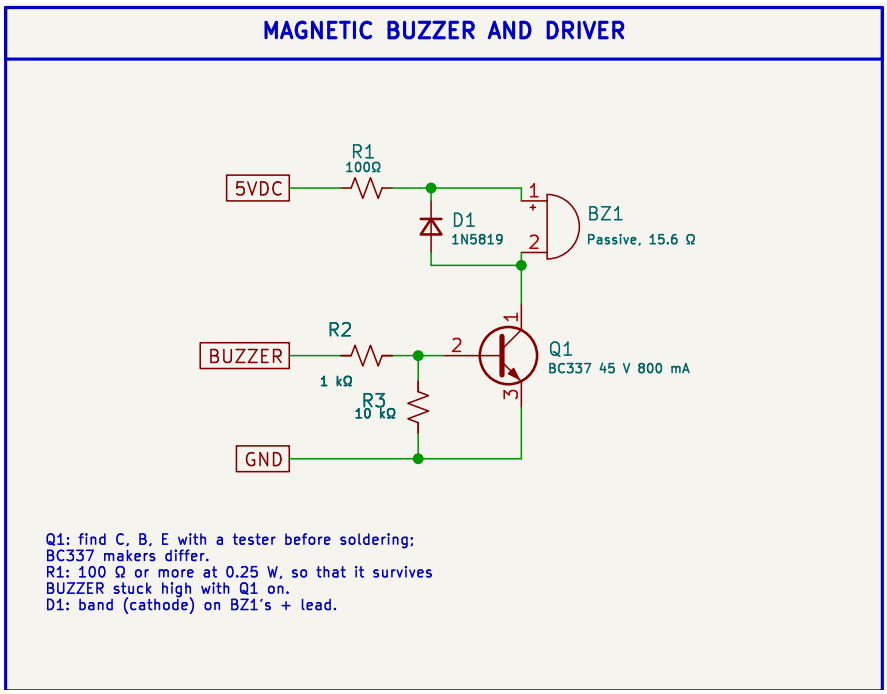


U3′s pins run GND, +, SW, DT, CLK (Figure 4). + is on `RP_3V3`. The module's own resistors pull CLK, DT and SW up to +: R1, R2 and R3 as printed on the

module. No pull-up is added outside it. A separate pull-up on the button was tried and removed, because the module's R3 already pulls SW up.

4.4 Buzzer and its driver

Figure 5. Schematic box MAGNETIC BUZZER AND DRIVER.



R1 runs from 5VDC to BZ1's + lead. BZ1's – lead goes to Q1's collector, and Q1's emitter to GND. D1 sits across BZ1 alone, cathode on BZ1's + lead. BUZZER reaches Q1's base through R2, and R3 runs from the base to GND (Figure 5). BUZZER high turns Q1 on and sounds BZ1.

- D1 carries the coil's current when Q1 turns off. It holds Q1's collector about one diode drop above BZ1's + lead, so the coil's turn-off spike cannot rise there.
- R3 holds Q1 off while GP8 does not drive BUZZER.
- R1, 100 Ω and 0.25 W or more, survives BUZZER stuck high with Q1 on.
- R2, 1 kΩ, lets GP8 drive Q1 with no pin-drive setting in the firmware.

5 Wiring

U1, the buzzer and its driver sit on one perfboard. The display cable and the encoder cable each have a JST XH connector at U1's end only, and are soldered to the module's pins at the other end (Figure 7). The USB-C extension joins J9 to P1. Wire every line by its signal name: no connector's pin 1 needs to be known.

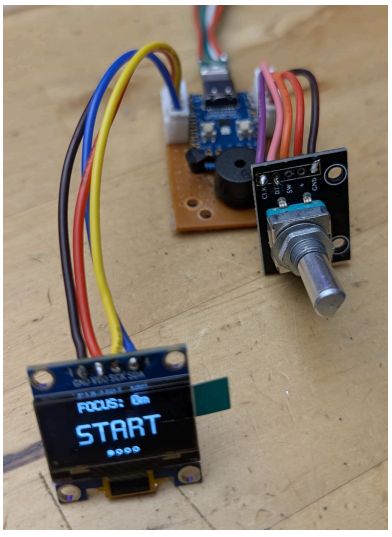


Figure 6. An earlier stage, out of the enclosure: the perfboard with U1 and the buzzer, the encoder, and the yellow-and-blue display later replaced.

JST XH: 2.5 mm pitch wire-to-board connectors, a header on the board and a housing crimped on the cable.

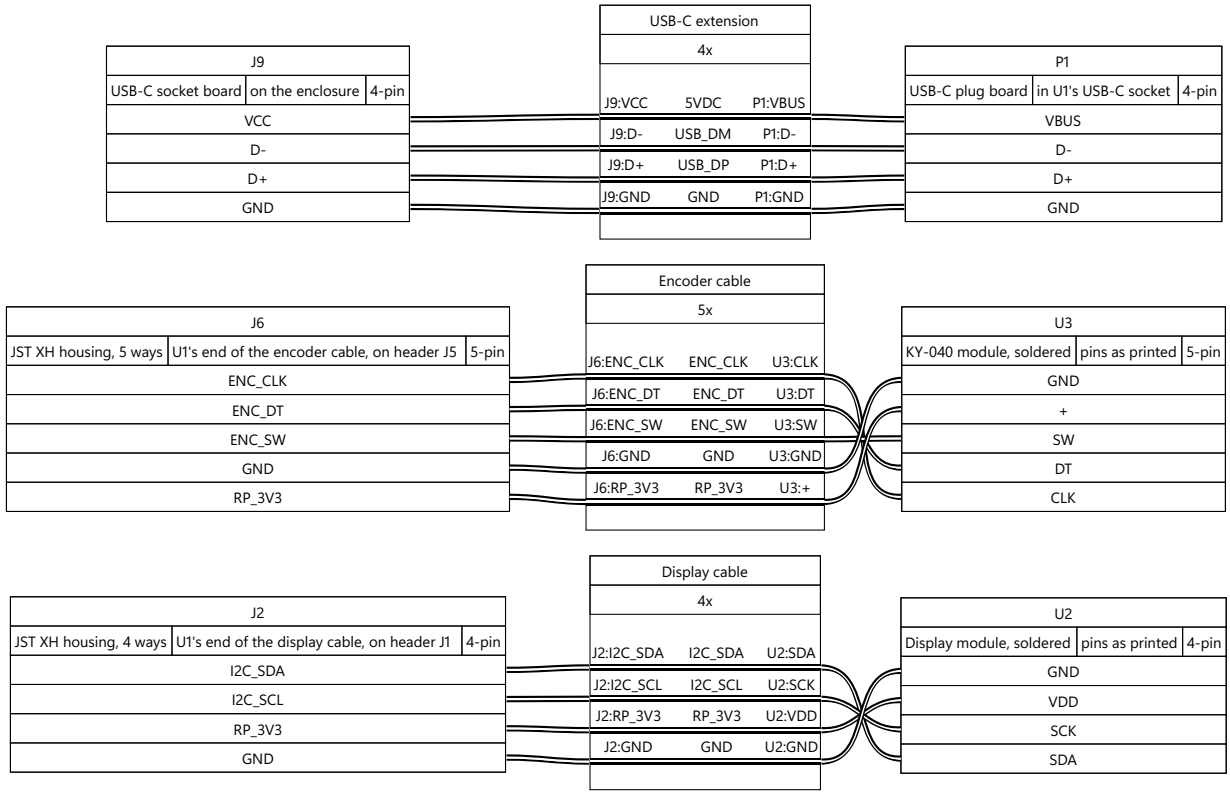
Table 5. Display cable and encoder cable, by signal.

Signal	Header	Housing	Module pin
GND	J1	J2	U2 GND
RP_3V3	J1	J2	U2 VDD
I2C_SCL	J1	J2	U2 SCK
I2C_SDA	J1	J2	U2 SDA
ENC_CLK	J5	J6	U3 CLK
ENC_DT	J5	J6	U3 DT
ENC_SW	J5	J6	U3 SW
GND	J5	J6	U3 GND
RP_3V3	J5	J6	U3 +

Table 6. Signal order at each connector. J1 and J2 are drawn mirrored; neither order is pin 1 to pin N.

Ref.	Where	Signals, top to bottom as the schematic draws them
J1	Header, perfboard	GND, RP_3V3, I2C_SCL, I2C_SDA
J2	Housing, display cable	I2C_SDA, I2C_SCL, RP_3V3, GND
J5	Header, perfboard	ENC_CLK, ENC_DT, ENC_SW, GND, RP_3V3
J6	Housing, encoder cable	ENC_CLK, ENC_DT, ENC_SW, GND, RP_3V3

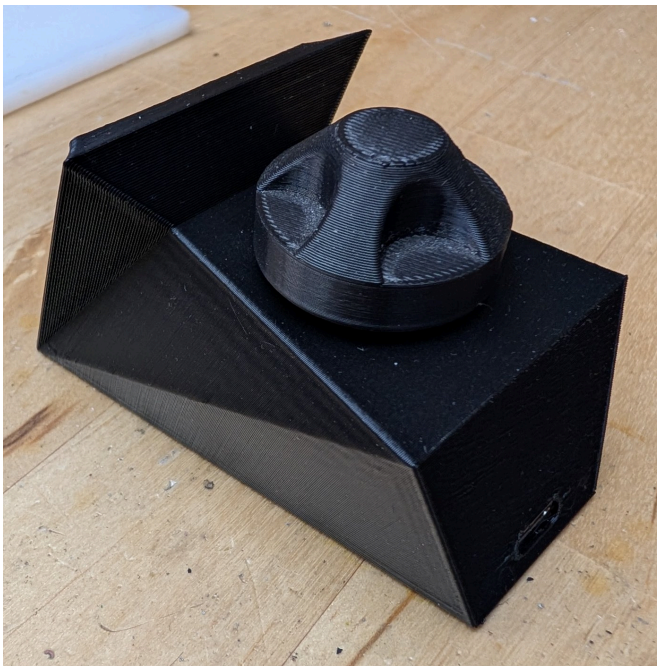
Figure 7. The three cables, by signal name. Each module end is soldered to the pin named.



NOTICE. J9′s VCC must reach P1′s VBUS, and J9′s GND P1′s GND. Find VBUS, GND, D+ and D- on your plug board before you solder it.

6 Enclosure

Figure 8. GERARD, back. J9, the USB-C socket, is in the lower corner.



Print `GERARD top.stl`, `GERARD bottom.stl` and `Knob FINAL.stl` in PLA, or any material that suits; no print setting is needed. Hold J9 and the perfboard in the enclosure with hot glue, or any fixing that suits (Figure 8). There are no screws or inserts.

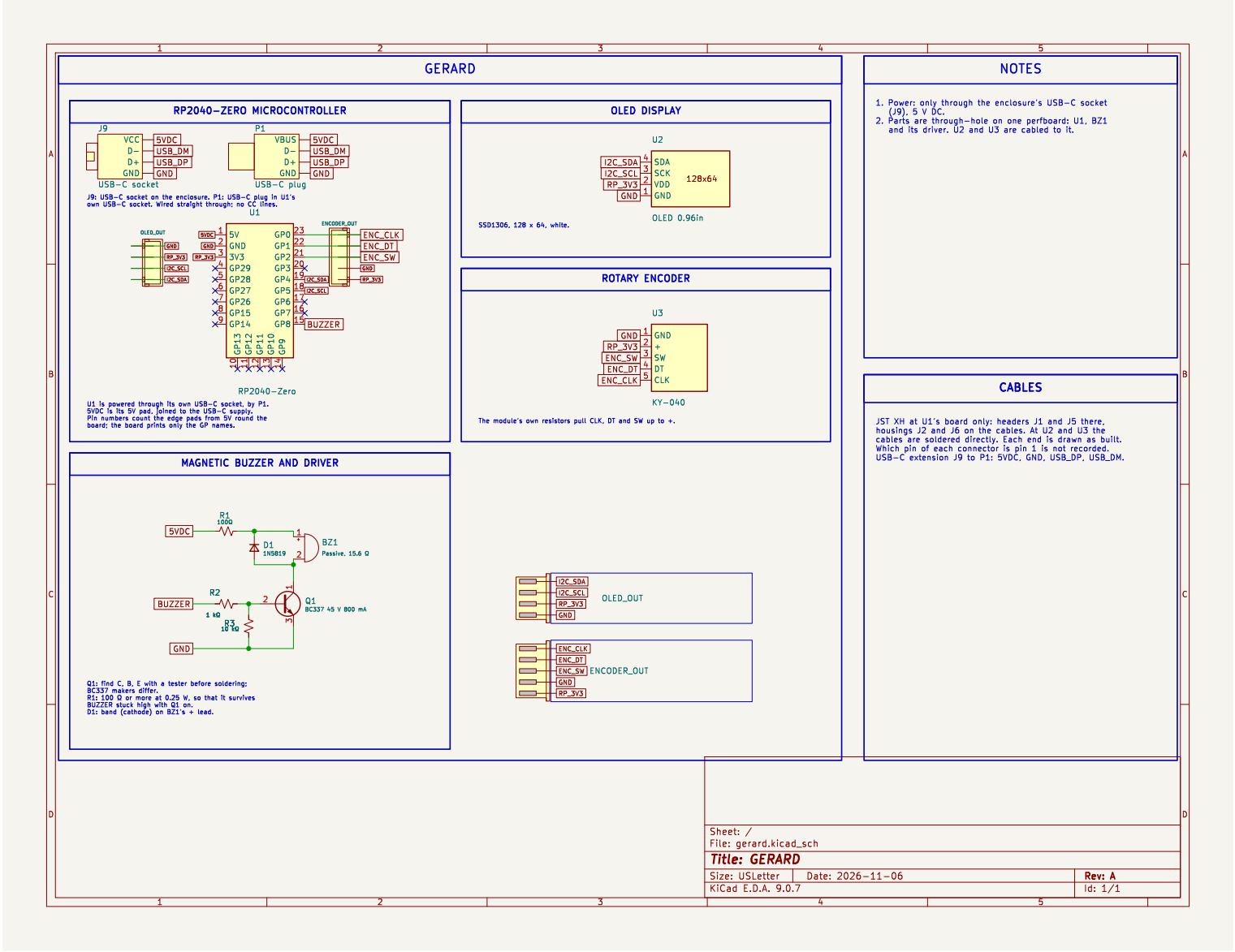
7 Datasheets

- RP2040 Datasheet, Raspberry Pi, build-date 2025-02-20, section 1.4.3, GPIO functions of GP4, GP5 and GP8: https://datasheets.raspberrypi.com/rp2040/rp2040-datasheet.pdf
- BC337, BC337-25, BC337-40 Amplifier Transistors, ON Semiconductor, BC337/D, Rev. 8, November 2013, page 1, Q1′s ratings. Current sheet: https://www.onsemi.com/pdf/datasheet/bc337-d.pdf
- Axial Lead Rectifiers 1N5817, 1N5818, 1N5819, onsemi, 1N5817/D, Rev. 11, August 2021, page 2, D1′s ratings: https://www.onsemi.com/pdf/datasheet/1n5817-d.pdf

The schematic was drawn with KiCad 9.0.7.

8 Schematic

Figure 9. The whole schematic, rev A.



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5VDC: 3
BUZZER: 4
BZ1: 1
D1: 2
ENC_CLK: 4
ENC_DT: 4
ENC_SW: 4
GND: 3
I2C_SCL: 4
I2C_SDA: 4
J1: 2
J2: 2
J5: 2
J6: 2
J9: 2
P1: 2
Q1: 1
R1: 2
R2: 2
R3: 2
RP_3V3: 3
U1: 1
U2: 1
U3: 1
USB_DM: 3
USB_DP: 3